D 11859 **(Pages : 4)** **Name.....**

Reg. No.....

THIRD SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2021

Mathematics

MAT 3B 03—CALCULUS AND ANALYTIC GEOMETRY

(2014—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A (Objective Type Questions)

Answer all **twelve** *questions.*
Each question carries 1 *mark.*

1. Find the eccentricity of the hyperbola $9x^2 - 16y^2 = 144$.

2. Find $\lim_{n\to\infty} \frac{4 - 7n^6}{n^6 + 3}$.

3. The inverse function of $\ln(x), x > 0$ is .....

4. If $f(x) = x^3 - 2$, find the value of $\frac{df^{-1}}{dx}$ at $x = f(2) = 6$.

5. Find a formula for the nth term of the sequence $1, -4, 9, -16, 25,$ .....

6. The centre-to-focus distance in the case of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is .....

7. Write a parametrization of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

8. $\lim_{n\to\infty} \frac{\ln n}{n}$ = ...

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9. Give an example of a conditionally convergence sequence.

10. Define a harmonic series.

11. If $\log_a x = \frac{\ln(x)}{k}$, then $k =$

12. The polar equation of the circle $x^2 + y^2 = 25$ is .....

(12 × 1 = 12 marks)

Part B (Short Answer Type)

Answer any **nine** *questions.*
Each question carries 2 *marks.*

13. Find the focus of the parabola $x^2 = 100\,y$.

14. Find $\lim_{x \to 0^+} x \cot x$.

15. Find $\lim_{n \to \infty} \frac{\cos n}{n}$.

16. State non-decreasing sequence theorem.

17. For what values of x do the power series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$ converge ?

18. Evaluate $\lim_{x \to 0} \frac{x - \sin x}{x^2}$.

19. Determine the conic section from the equation $3x^2 - 6xy + 3y^2 + 2x - 7 = 0$.

20. Identify the geometric figure in the cartesian plane represented by the polar equation $r \cos(\theta - \pi/3) = 2$.

21. Test the convergence of $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$.

22. For what values of x do $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$ converge ?

23. Write the formula for finding the area using polar co-ordinates.

24. Find the polar equation of the hyperbola with eccentricity $3/2$ and directrix at $x = 2$.

(9 × 2 = 18 marks)

Part C (Short Essay Type)

Answer any **six** *questions.*
Each question carries 5 *marks.*

25. Using the deflation of natural logarithm, prove that $\ln(xy) = \ln x + \ln y$.

26. Evaluate $\lim_{x \to 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$.

27. Find a formula for the nth partial sum of the series $2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + ... + \frac{2}{3^{n-1}}$ and use it to find the serie's sum if the series converges.

28. Use integral test to show that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for $p > 1$.

29. Show that the point $(2, 3\pi/4)$ lies on the curve $r = 2 \sin 2\theta$.

30. Find the Maclaurin series for the function $f(x) = \sinh x$.

31. Find the radius and interval of convergence of the series $\sum_{n=0}^{\infty} (-1)^n (4x+1)^n$.

32. Calculate the value of e with an error less than 10^{-6}.

33. Draw the curve represented by the parametric equations $x = \cos t, y = -\sin t, t \in [0, \pi]$.

(6 × 5 = 30 marks)

Turn over

Part D (Essay Type)

Answer any **two** *questions.*
Each question carries 10 *marks.*

34. Find the area of the surface generated by revolving the right handed loop of the lemniscate $r^2 = \cos 2\theta$ about y-axis.

35. Find the Taylor series generated by $f(x) = 1/x$ at $a = 2$. Where if anywhere, does the series converges to $1/x$?

36. a) The x and y axes are rotated through an angle of $\pi/4$ radians about the origin. Find the converted equation of the hyperbola $2xy = 9$ in the new co-ordinates.

b) Evaluate $\lim_{x \to \left(\frac{\pi}{2}\right)^{-1}} \frac{\sec x}{1 + \tan x}$.

(2 × 10 = 20 marks)